VIRTUE EPISTEMOLOGY

A Study Pack for A-Level Philosophy Students

Based on Sosa's 'The Raft and the Pyramid' (1980) and the Stanford Encyclopedia of Philosophy

Part One: What Is Virtue Epistemology?

The Problem Sosa Is Responding To

Ernest Sosa's 1980 paper 'The Raft and the Pyramid: Coherence versus Foundations in the Theory of Knowledge' is one of the most important texts in twentieth-century epistemology. To understand it — and the broader tradition of virtue epistemology it helped to inspire — we first need to understand the problem it addresses.

For centuries, philosophers have wrestled with a fundamental question: what gives a belief its justification? Two great competing answers have dominated the debate. Foundationalism holds that our system of beliefs is like a pyramid: at the bottom sit a small number of basic beliefs that are self-evident or indubitable, and all other justified beliefs rest on these foundations. Coherentism rejects this picture in favour of a raft: there is no special foundation, and every belief is justified by how well it coheres with the web of other beliefs one holds. Each plank on the raft supports the others; none is untouchable.

Sosa's paper exposes deep problems with both views and gestures toward a third path: one grounded not in the structure of beliefs, but in the virtues and reliable capacities of the knower.

The Foundationalist Pyramid — and Its Weakness

Foundationalism, in its classical form associated with Descartes, holds that every justified belief must ultimately rest on beliefs that are infallible and indubitable — beliefs about which the thinker simply cannot be wrong. From these bedrock certainties, all other knowledge is derived by deductive proof.

Sosa identifies a fatal twofold weakness here. First, there are not nearly as many indubitable truths as Descartes supposed. Second, even if we grant a small core of certainties, very little of our everyday knowledge — of the external world, of other people, of history — can be deduced from them. If we hold on to radical foundationalism, Sosa argues, we must admit that we know far less than we ordinarily suppose. As a committed particularist — someone who begins from the conviction that we do in fact know a great deal — Sosa refuses to accept this sceptical conclusion. Instead, he rejects the criteria and looks for a better account.

The Coherentist Raft — and Its Weakness

Coherentism, by contrast, holds that a belief is justified when it coheres within a comprehensive and consistent system of beliefs. It abandons the metaphor of the pyramid in favour of Neurath's raft: floating freely, no single plank is the unalterable foundation, but the whole structure holds together.

Sosa's most devastating objection to coherentism is what we might call the detachment problem. Coherentism makes justification entirely a matter of relations among beliefs — it is blind to how those beliefs connect to the world. But consider: if I have a splitting headache, my belief that I am in pain is surely justified by the headache itself, not merely by how it coheres with my other beliefs. We can imagine an alternative system of beliefs, coherent in its own right, that includes the belief that I am not in pain — and coherentism cannot explain why the first system is better than the second. The actual headache is not itself a belief; it is something in the world. And that shows that justification cannot be solely a matter of relations among beliefs.

Sosa's Core Insight: The Raft and Pyramid Both Fail

Radical foundationalism fails because it makes our stock of knowledge impossibly small.

Coherentism fails because it severs the connection between our beliefs and reality.

Both are varieties of what Sosa calls 'formal foundationalism' — attempts to give a systematic, non-epistemic account of how justification arises. The question is: what can do better?

The Regress Problem

Sosa also carefully analyses the regress of justification. If every belief requires justification from another belief, we seem condemned to an infinite chain: B1 is justified by B2, which is justified by B3, and so on without end. Foundationalists use this regress argument as their strongest weapon, concluding that the chain must terminate in non-inferentially justified beliefs — the foundations.

Sosa's treatment of this argument is nuanced. He distinguishes between potential regresses (where each belief could justify the next, but none is actually justified) and actual regresses (where each belief is genuinely justified by its successors). He does not simply accept that a non-terminating regress is impossible in principle. What matters is not the mere possibility of an infinite regress, but the positive question of what makes any belief actually justified.

The Move Toward Intellectual Virtue

In the final and most influential section of the paper, Sosa sketches what he calls a 'reliabilism': an epistemology of intellectual virtues. The key move is to shift attention from the structure of individual beliefs to the stable, reliable dispositions of the knower.

Sosa draws an analogy with ethics. In moral philosophy, we can distinguish between primary justification — which applies to virtues and stable dispositions — and secondary justification — which applies to particular acts in virtue of their source in those virtues. A compassionate action is right partly because it flows from compassion, a virtuous character trait.

Sosa proposes an exactly parallel structure for epistemology. Primary justification applies to intellectual virtues: stable cognitive dispositions — such as keen perception, reliable memory, sound reasoning, and careful introspection — that consistently lead a person toward true beliefs. Secondary justification then applies to individual beliefs, which are justified in virtue of their arising from these intellectual virtues.

Sosa's Positive Proposal in Brief

Intellectual virtues are stable dispositions for belief acquisition that reliably tend toward

truth.
A belief is justified (secondarily) when it arises from such a virtue.
The virtues themselves are justified (primarily) by their reliable contribution to truth-acquisition.
This connects justification to the actual cognitive character of the knower, not merely to the logical relations among beliefs.

Why This Is a Breakthrough

Sosa's virtue-based approach breaks the deadlock between foundationalism and coherentism by grounding justification in something external to the structure of beliefs — the reliable functioning of the knower's cognitive faculties. This makes it a form of externalism: the factors that make a belief justified need not be ones the believer is consciously aware of. What matters is not whether you can articulate your reasons, but whether your belief arose from a reliably truth-tracking faculty.

Crucially, Sosa also insists that intellectual virtues must take account not only of the subject's intrinsic nature but also of their environment and epistemic community. A faculty counts as a virtue partly because of the conditions in which it operates. This environmental sensitivity distinguishes Sosa's view from naive process reliabilism and opens the door to the richer virtue epistemology that he and others would develop in subsequent decades.

Sosa's Lasting Legacy

Contemporary virtue epistemology, which flourished from the 1980s onwards, takes Sosa's seminal insight as its starting point. Two broad camps emerged. Virtue reliabilists — including Sosa himself, John Greco, and Duncan Pritchard — define intellectual virtues in terms of reliable cognitive faculties such as perception, memory, and introspection. Virtue responsibilists — such as Lorraine Code, Linda Zagzebski, and Heather Battaly — broaden the notion to include cultivated character traits: intellectual courage, intellectual humility, and open-mindedness. Together these approaches have reinvigorated epistemology, addressing not just knowledge and justification but understanding, wisdom, epistemic value, and the social dimensions of cognition.

Part Two: Objections and Responses to Virtue Epistemology

The Stanford Encyclopedia of Philosophy (Turri, Alfano, and Greco, 2021) surveys a rich landscape of challenges to virtue epistemology. Here we examine the most important, grouped thematically.

1. The Gettier Problem

The Objection

Edmund Gettier (1963) showed that justified true belief is not sufficient for knowledge. His cases follow a recipe: start with a justified belief that would normally ground knowledge; add

a dose of bad luck that would prevent the belief being true; then add compensating good luck that makes it true anyway. The VE response is that knowledge requires you to believe the truth because of your intellectual virtue — Gettier subjects fail this test because their true belief is not explained by their virtue but by luck.

The Problem: Fake Barn Country

A particularly awkward case is the 'fake barn' scenario. Henry drives through countryside filled with convincing fake barns and happens to look at the one real barn in the area. He forms a true belief based on his excellent visual competence. Yet intuitively, many philosophers say he doesn't know — he just got lucky. Sosa's account seems to imply he does know, since his belief is correct because of his perceptual competence. Critics like Pritchard argue this is a counterexample to virtue reliabilism.

Responses

Virtue epistemologists have responded in two ways. Some (like Jarvis and Littlejohn) argue that fake barn cases actually do count as apt belief and that the intution that Henry doesn't know should be revised. Others, notably Pritchard, combine virtue reliabilism with an additional safety condition: a belief counts as knowledge only if it couldn't easily have been held falsely. Since Henry could very easily have looked at a fake barn and formed a false belief, his belief is unsafe and hence not knowledge. Experimental work (Colaço, Buckwalter, Stich, and Machery, 2014) has shown that ordinary people actually tend to attribute knowledge in fake barn cases — suggesting the intuition may be less firm than philosophers assumed.

2. The Credit Thesis and the Problem of Testimony

The Objection

A central claim in virtue epistemology is the credit thesis: you know only if you deserve credit for believing the truth. Jennifer Lackey (2007) challenges this with the case of 'Chicago Visitor'. Morris arrives at a train station and asks a passerby for directions to the Sears Tower. The passerby gives perfect directions based on exceptional local knowledge. Morris forms a true belief. Lackey argues Morris clearly knows where the tower is — but his reliable cognitive faculties play only a minor role in the explanation. Most of the credit belongs to the passerby. So the credit thesis is false.

Responses

Sosa responds that Morris still deserves partial credit: his 'socially seated competence' — the ability to identify reliable informants and accept testimony appropriately — is itself an intellectual virtue. Greco argues that cooperative success can distribute credit to multiple participants, even those who contribute less. Riggs distinguishes praiseworthiness from attributability: knowledge requires the true belief to be attributable to you as an agent, but not that you be fully praiseworthy for it. Lackey's counter-dilemma is sharp: either the credit requirement is strong enough to rule out Gettier subjects, in which case it rules out too much testimonial knowledge; or it is weak enough to accommodate testimonial knowledge, in which case Gettier problems reassert themselves.

3. Epistemic Situationism

The Objection

Virtue epistemology assumes people possess stable intellectual virtues — reliable, enduring cognitive dispositions. But empirical psychology raises a serious challenge. Research on heuristics and cognitive biases (the availability heuristic, confirmation bias, the

representativeness heuristic) suggests that human cognition is highly sensitive to seemingly irrelevant situational factors — mood, social pressure, framing effects. Mark Alfano (2012) formulated this as an inconsistent triad: we cannot simultaneously hold that (a) most humans have a great deal of knowledge, (b) most humans' intellectual dispositions are not genuine virtues because they are too situationally variable, and (c) knowledge requires manifesting intellectual virtue.

Responses

Four main responses have emerged. First, some researchers point to more heartening empirical evidence: heuristics, when properly monitored by reflective, 'System 2' thinking, can be reliable — and this monitoring itself constitutes a virtue such as intellectual humility. Second, conciliatory approaches suggest VE should focus less on achieving virtue and more on recognising and overcoming vice — intellectual humility becomes a matter of knowing one's cognitive limits. Third, extended cognition approaches argue that virtues need not be purely internal: by structuring our informational environment wisely (choosing reliable news sources, cultivating good epistemic communities), we can offload cognitive reliability onto external scaffolding. Fourth, Turri (2017) argues there is simply no strong empirical or conceptual evidence that knowledge requires the kind of cross-situational reliability that epistemic situationism challenges.

4. The Value Problem

The Objection

Plato's Meno posed the question: why is knowledge more valuable than mere true belief, if both serve equally well for guiding action? Zagzebski (2003) argues reliabilism cannot answer this: a true belief produced by a reliable process is no more valuable than the same true belief produced by lucky guessing, just as a good cup of coffee is not made better simply because it was brewed by a reliable coffee machine. The reliability adds no extra value.

The VE Solution

Virtue epistemology offers a distinctive solution. Greco, Sosa, and Riggs argue that knowledge is an achievement — success through the exercise of ability. Achievements carry a distinctive kind of value over and above their outcomes. A runner who wins a race through training and skill achieves something more valuable than one who wins by accident, even if both cross the finish line first. Similarly, believing the truth because of your intellectual virtue is more valuable than luckily stumbling on the truth. VE thus introduces a new vector of value — credit-worthiness — that is absent from true belief formed accidentally.

5. The Virtue Reliabilist / Responsibilist Divide

The Tension

A persistent internal tension in VE concerns which capacities count as intellectual virtues. Virtue reliabilists (Sosa, Greco) focus on reliable cognitive faculties: good eyesight, accurate memory, sound perceptual discrimination. Virtue responsibilists (Zagzebski, Code, Battaly) argue these faculty-virtues miss the distinctively ethical dimension of epistemic virtue — traits like intellectual courage, intellectual humility, and conscientiousness require effort, motivation, and responsibility.

Synthesis

A growing consensus holds that both faculty-virtues and trait-virtues are needed for a complete epistemology. Faculty-virtues seem indispensable for explaining basic perceptual

and memorial knowledge; trait-virtues are needed to account for richer intellectual achievements like wisdom, understanding, and the courage to pursue truth in the face of social pressure. Some philosophers, such as Baehr, argue that trait-virtues are actually required even for knowledge in certain cases — intellectual courage and perseverance may figure centrally in how a knower arrived at the truth.

6. Contextualism

John Greco has developed 'virtue contextualism', arguing that knowledge attributions are context-sensitive in the same way that explanatory talk generally is. Whether we say someone believes the truth because of their virtue depends on what features of the situation are salient and what our explanatory purposes are. This is meant to handle cases where knowledge is difficult to attribute — such as sceptical scenarios — without abandoning our ordinary practices of knowledge attribution. Critics argue this approach is unmotivated and that experimental evidence does not support the predicted context-sensitivity of knowledge attributions.

Glossary of Technical Terms

The following terms are central to understanding virtue epistemology and the Sosa text. Master them before attempting the essay questions.

Term	Definition
AAA-model	Sosa's framework for assessing performances as accurate (achieving the aim), adroit (manifesting competence), and apt (accurate because adroit). Knowledge is apt belief.
Achievement	Success that results from the exercise of ability or competence, rather than from luck. VE holds that knowledge is a cognitive achievement.
Anti-scepticism	The view that humans do in fact possess a great deal of knowledge — the starting point of particularism, and a key commitment of most virtue epistemologists.
Apt belief	In Sosa's framework, a belief that is true because it manifests the believer's epistemic competence. Apt belief is identified with knowledge.
Coherentism	The view that a belief is justified when it coheres with — fits consistently and explanatorily within — a comprehensive system of beliefs. Associated with the metaphor of Neurath's raft.
Credit thesis	The claim that knowledge is a credit-worthy state of the agent — you know only if you deserve credit for believing the truth. Central to VE accounts of epistemic value.
Epistemic community	The social group within which a knower is embedded, whose shared practices, norms, and testimony shape individual knowledge and justification.

Term	Definition
Epistemic injustice	A wrong done to someone in their capacity as a knower. Miranda Fricker identifies two main varieties: testimonial injustice and hermeneutical injustice.
Epistemic situationism	The challenge to VE from cognitive psychology, arguing that human intellectual dispositions are too sensitive to situational factors to count as genuine virtues.
Externalism	The view that the factors that determine whether a belief is justified (or counts as knowledge) can be external to the believer's conscious awareness.
Faculty virtue	A reliable cognitive faculty — such as perception, memory, or introspection — understood as an intellectual virtue. Favoured by virtue reliabilists like Sosa and Greco.
Fake barn scenario	A thought experiment in which a subject is in an area full of fake barns but happens to look at the one real barn. Used to test virtue epistemological accounts of knowledge.
Formal foundationalism	Sosa's term for any view that specifies, in non-epistemic terms, the conditions under which epistemic justification supervenes on non-epistemic properties.
Foundationalism	The view that justified beliefs form a pyramid, with a special class of basic beliefs at the base that are self-evident or indubitable, and all other justified beliefs resting on them.
Gettier problem	The problem of explaining why justified true belief is not sufficient for knowledge, following Edmund Gettier's 1963 cases showing that someone can have JTB without knowing.
Hermeneutical injustice	A form of epistemic injustice in which people are denied the conceptual resources to make sense of and communicate their own experiences.
Intellectual humility	An intellectual virtue involving appropriate recognition of one's cognitive limitations, openness to evidence, and resistance to overconfidence.
Intellectual virtue	A stable cognitive trait that promotes intellectual flourishing or makes for an excellent cognizer. VE divides these into faculty-virtues and trait-virtues.
Particularism	The epistemological approach that begins from the conviction that we do know specific things, and derives criteria from that, rather than first fixing criteria and then asking what can be known.
Primary justification	In Sosa's framework, justification that applies directly to intellectual virtues or stable dispositions, through their contribution to truth.
Radical foundationalism	The strict Cartesian version of foundationalism holding that basic beliefs must be infallible and indubitable. Sosa argues this leads to scepticism.
Regress of justification	The chain of justification (B1 justified by B2, B2 by B3, etc.) that, if allowed to continue infinitely, motivates the foundationalist's demand for a stopping point.
Safety condition	The requirement that knowledge must not only be true and formed via a competence, but that it couldn't easily have been false. Used by Pritchard to supplement virtue reliabilism.

Term	Definition
Secondary justification	In Sosa's framework, justification that attaches to individual beliefs in virtue of their origin in an intellectually virtuous disposition.
Supervenience	The relation whereby epistemic properties (justification, knowledge) depend on, and cannot vary independently of, non-epistemic properties (the causal history, reliability, content of belief).
Testimonial injustice	A form of epistemic injustice in which someone's testimony is accorded less credibility than it deserves because of prejudice related to their identity.
Trait virtue	A cultivated intellectual character trait — such as intellectual courage, humility, or open-mindedness — understood as an intellectual virtue. Favoured by virtue responsibilists.
Value problem	The problem of explaining why knowledge is more valuable than mere true belief, if both serve equally well for guiding action.
Virtue contextualism	Greco's view that knowledge attributions are context-sensitive, because the 'because of your virtue' relation in the definition of knowledge depends on what is explanatorily salient in context.
Virtue reliabilism	The branch of VE that understands intellectual virtues as reliable cognitive faculties. Prominent figures include Sosa, Greco, and Pritchard.
Virtue responsibilism	The branch of VE that understands intellectual virtues as cultivated, responsible character traits. Prominent figures include Zagzebski, Code, and Battaly.

Comprehension and Essay Questions on Sosa's 'The Raft and the Pyramid'

Read Sosa's article carefully before attempting these questions. Focus on precision and always use examples to support your answers.

Section A: Understanding the Text

These questions test your comprehension of Sosa's argument. Aim for clear, concise answers grounded directly in the text.

Q1. What are Sosa's two starting assumptions (A1 and A2) about knowledge and justification? Why does he make these assumptions, and what work do they do in the argument?

Q2. Explain the metaphors of 'the raft' and 'the pyramid'. Which epistemological positions do they represent, and what does each metaphor capture about those positions?

Q3. What is 'radical foundationalism', and what does Sosa identify as its 'fatal weakness'? Why does he think this weakness forces us to reject Cartesian foundationalism?

Q4. Reconstruct the coherentist's dilemma argument against foundationalism (pages 6–7 of the article). What are its two lemmas and its conclusion? Why does Sosa think it fails?

Q5. What is the 'headache counterexample' to coherentism (section 9)? Explain in your own words what the example shows and why it is damaging to the coherentist position.

Q6. Sosa distinguishes between 'formal foundationalism' and 'substantive foundationalism'. Explain this distinction. Which of the two positions — coherentism and substantive foundationalism — does Sosa think is superior, and why?

Q7. In section 11, Sosa sketches his positive proposal — 'reliabilism: an ethics of moral virtues and an epistemology of intellectual virtues'. Explain the analogy with ethics that he uses and what the proposal amounts to for epistemology.

Section B: Critical Engagement

These questions ask you to evaluate the arguments critically. Draw on both the text and the Stanford Encyclopedia material.

Q8. Sosa argues that both foundationalism and coherentism aim at 'formal foundationalism of the highest grade'. What does he mean by this? Do you think this alleged kinship between the two positions is philosophically significant?

Q9. The 'doxastic ascent arguments' (section 8) are presented as attacks on foundationalism. Sosa argues they fail and that they apply equally to coherentism. Do you agree? Are these arguments more damaging to one position than the other?

Q10. Consider the fake barn case. Does Henry know that the barn in front of him is a barn? How would Sosa's virtue epistemology answer this question? Do you find that answer satisfying?

Q11. Jennifer Lackey's 'Chicago Visitor' case challenges the credit thesis. Set out the argument carefully. Which of the virtue epistemologist responses (Sosa, Greco, or Riggs) do you find most convincing, and why?

Section C: Essay Questions

Choose one of the following essay questions. Plan your response before you write. Aim for 700–900 words with a clear argument.

Essay 1. 'Sosa's shift from foundationalism and coherentism to virtue epistemology is a genuine philosophical advance, not merely a change of subject.' Discuss with reference to 'The Raft and the Pyramid'.

Essay 2. 'Virtue epistemology solves the Gettier problem and the value problem but fails in the face of Lackey's testimony cases and epistemic situationism.' Critically assess this claim.

Essay 3. 'Virtue reliabilism and virtue responsibilism are not genuine rivals but complementary approaches that a complete epistemology must incorporate.' Do you agree?

End of Study Pack